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MAE 3240 HW 2

1.

a)

$$q = hA_{side}(T_s - T_\infty) + hA_{top}(T_s - T_\infty)$$

$$q = 30(15)(2\pi r h + \pi r^2)$$

$$\boxed{q = 1767.15\,W}$$

b) $q_r = -kA\frac{dT}{dr} = -k(2\pi r L)\frac{\partial T}{\partial r}$

$$T(r) = \frac{T_{s_1} - T_{s_2}}{\ln(\cancel{r_1/r_2})}\ln(\cancel{r/r_2}) + T_{s_2}$$

$$R_{conduction} = \frac{\ln(1 + \frac{L}{R})}{2\pi k H} = 0.0624$$

$$R_{convection} = \frac{1}{h(2\pi(.52)h)} = 0.0204$$

$$q_{side} = \frac{30}{.0204 + .0624} = 362.32\,W$$

side

$$R_{convection} = \frac{1}{h(\pi r^2)} = 0.0785$$

$$R_{conduction} = \frac{L}{\pi r^2 k} = 0.29469$$

$$q_{top} = \frac{30}{.0785 + .29469} = 90.04\,W$$

top

$$q_{top} + q_{side} = 362.3 + 90.04 \approx \boxed{452\,W}$$

2.

a) $q_{in} = q_{out}$

$$T(L) = h\left[T(L) - T_\infty\right]$$

$$T_L = T_\infty \frac{q''}{h}$$

$$T_L = 20 + 10 = 30^\circ C$$

$$\boxed{T_L = 30^\circ C}$$

$$T(0) = \frac{q''L}{k} + T_L$$

Conduction

$$T(0) = 100 + 30 = 130°C$$

b)

$$\frac{q''L}{k} + q''h + T_\infty = 200$$

$$180 = q''\left(\frac{L}{k} + \frac{1}{h}\right)$$

$$q'' = \frac{180}{\left(\frac{L}{k} + \frac{1}{h}\right)} = \boxed{1.64 \text{ kW/m}^2}$$

c) - thinner heater

- change coolant to account for higher h

- more thermally conductive material (increase k)

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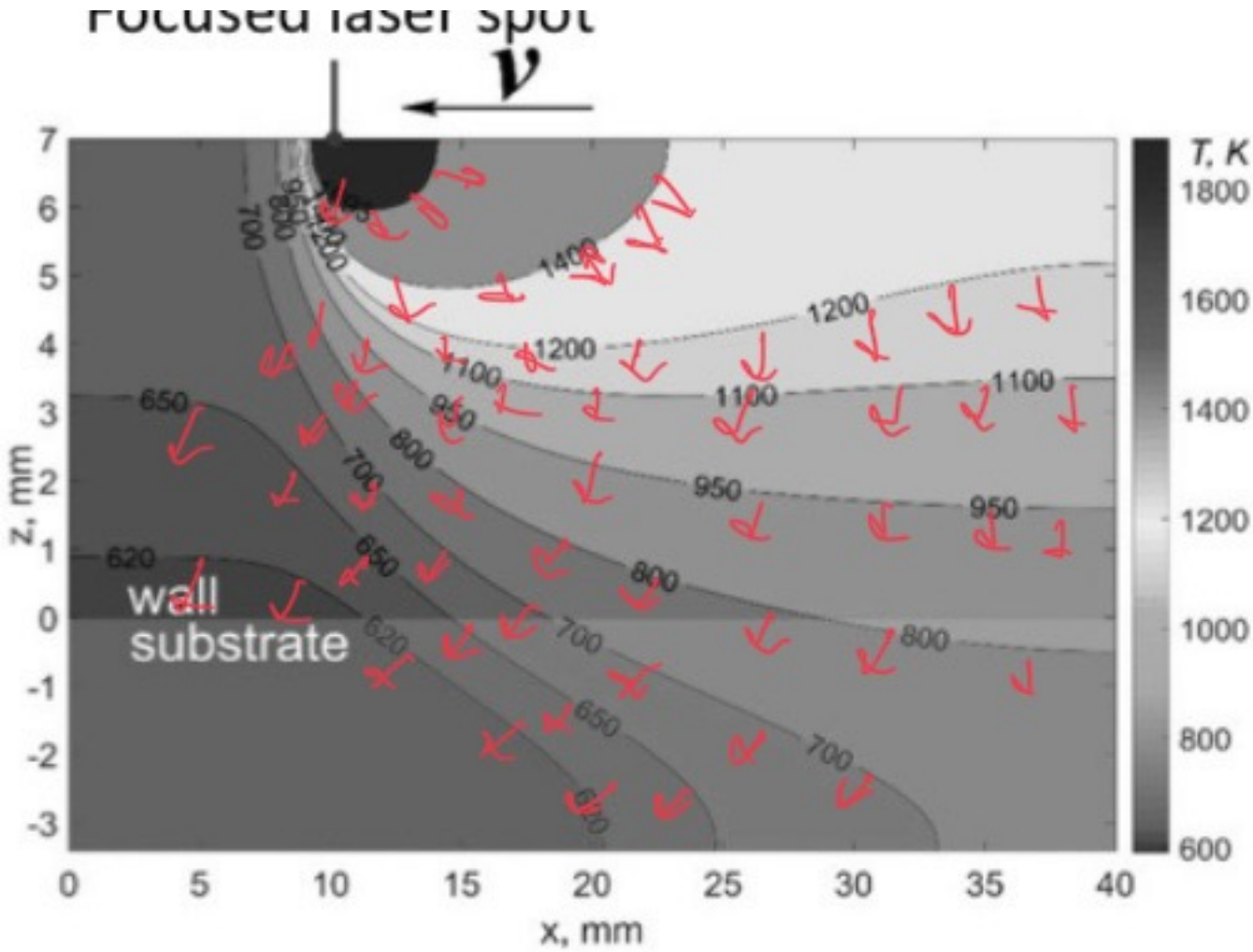
3. a)



b)

the left side's flux is larger due to the higher density of isothermic liner